

Assignment 1: Building a Modern ELT Pipeline

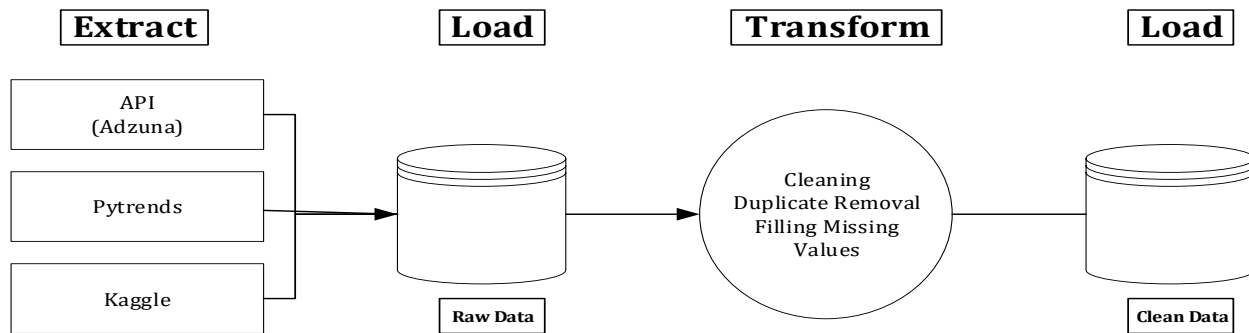
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GitHub Repository

<https://github.com/HasanAmjad/AI620-ASSIGN1-AI-JOB-MARKETS-ELT-PIPELINE>

Project Pipeline



Summary

This project focused on the **AI Labor Markets** domain, analyzing trends in AI-related job postings, salary distributions, required skills, and public search interest. Three primary data sources were used: the Adzuna Jobs API, a Kaggle AI jobs dataset, and Google Trends.

Key Findings and Analysis

- AI-related search interest has shown a rising trend in 2025–2026.
- Monthly and weekly job posting aggregation from Adzuna demonstrated measurable hiring activity variation over time.
- Principal AI and advanced engineering roles commanded the highest compensation, followed by senior data science positions.
- A weak positive correlation ($r \approx 0.28$) was observed between the number of required skills and salary.

Challenges Encountered

- Reddit API had access restrictions, so I switched to the Adzuna API instead.
- The data was largely complete, but some fields had empty strings instead of proper null values, so I had to handle and standardize them.
- Salary and date fields were originally stored as strings, so I converted them to numeric for proper analysis.
- The API data was limited since only a few pages were extracted, which restricted deeper statistical analysis but was sufficient for exploratory insights.